

TUROCHAK, Russian Federation

WMO: 360610

Lat: 52.250N

Lon: 87.127E

Elev: 1071

StdP: 14.14

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.8	-24.6	-35.2	0.8	-30.6	-28.9	1.1	-24.5	13.2	23.5	9.9	13.7	0.1	210	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	87.2	66.7	84.0	65.4	80.7	63.9	69.2	82.4	67.5	79.6	66.0	77.0	4.2	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.1	96.7	73.0	63.4	91.0	71.5	62.0	86.3	70.3	34.0	82.8	32.7	79.7	31.4	77.2	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.4	9.7	8.0	-35.9	94.2	6.3	3.2	-40.5	96.5	-44.2	98.4	-47.7	100.2	-52.3	102.5
			-35.8	72.6	5.6	2.4	-39.9	74.3	-43.2	75.7	-46.4	77.0	-50.5	78.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.7	1.2	9.1	24.7	40.1	52.7	63.3	66.9	63.0	51.8	38.7	20.5	6.3
	DBStd	24.63	12.45	11.92	11.06	9.48	8.26	6.36	5.17	6.00	7.36	8.46	12.68	12.39
	HDD50	6495	1512	1144	785	315	69	2	0	1	66	360	885	1356
	HDD65	10548	1977	1564	1250	748	387	103	40	109	399	815	1335	1821
	CDD50	1629	0	0	0	17	153	402	523	404	120	10	0	0
	CDD65	207	0	0	0	0	7	52	98	47	3	0	0	0
	CDH74	2770	0	0	0	19	241	801	1019	608	81	1	0	0
	CDH80	794	0	0	0	2	56	227	320	174	15	0	0	0
Wind	WSAvg	2.5	1.2	1.4	2.4	3.4	3.6	3.0	2.7	2.7	2.8	2.7	2.3	1.5
Precipitation	PrecAvg	32.3	1.4	1.3	1.5	2.7	3.2	3.5	3.8	3.6	3.1	2.7	2.8	2.6
	PrecMax	38.6	2.6	2.2	3.1	5.9	6.6	5.8	7.0	7.8	6.3	5.7	5.5	7.0
	PrecMin	23.3	0.4	0.3	0.3	0.9	1.7	1.5	2.1	1.8	1.1	0.6	1.2	0.6
	PrecStd	3.7	0.6	0.5	0.7	1.5	1.3	1.3	1.2	1.5	1.5	1.2	1.3	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.7	39.2	56.0	76.4	86.0	89.8	91.6	91.5	82.1	68.8	50.0	33.6
		MCWB	31.2	32.9	43.2	53.9	62.0	66.1	69.4	68.0	62.8	53.4	42.5	32.5
	2%	DB	27.3	34.3	50.6	68.5	80.8	85.8	87.2	84.9	75.5	62.0	43.3	29.8
		MCWB	26.1	30.3	39.9	50.7	58.4	65.2	68.1	66.9	59.9	50.0	38.3	28.7
	5%	DB	23.4	30.5	45.4	62.7	76.3	82.4	83.8	80.6	70.4	56.2	38.5	27.2
		MCWB	22.3	27.6	37.1	48.5	56.6	63.8	66.9	65.2	57.4	47.6	35.8	26.2
	10%	DB	19.5	26.8	40.5	57.1	71.0	78.8	80.2	76.8	65.9	51.2	34.8	23.9
		MCWB	18.5	24.3	34.3	45.6	55.2	62.9	65.5	63.7	55.3	45.2	33.2	23.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.8	33.7	44.3	55.7	64.1	69.4	72.3	70.9	64.1	54.6	43.5	32.8
		MCDB	32.7	37.3	54.9	72.1	81.8	83.8	85.1	86.2	79.1	66.2	49.0	33.7
	2%	WB	26.2	30.8	40.4	52.3	61.1	67.3	70.0	68.1	60.9	51.4	38.8	29.0
		MCDB	27.1	33.7	48.8	65.1	75.1	80.7	82.9	82.2	73.0	59.7	42.6	29.8
	5%	WB	22.4	27.6	37.7	49.7	58.7	65.7	68.3	66.2	58.5	48.4	35.8	26.3
		MCDB	23.5	30.5	44.6	60.0	71.1	77.9	79.8	77.9	68.5	55.1	38.2	27.2
	10%	WB	18.6	24.3	34.9	46.6	56.4	64.2	66.8	64.5	56.2	45.4	33.3	23.1
		MCDB	19.6	26.6	39.7	56.4	67.5	75.2	77.3	74.4	64.2	50.9	34.8	24.0
Mean Daily Temperature Range	5% DB	MDBR	20.9	26.7	26.4	22.8	24.3	23.1	20.8	21.4	22.1	18.1	15.4	17.5
		MCDBR	17.0	27.4	32.1	34.9	36.4	31.9	29.2	30.7	32.5	30.1	16.9	14.0
		MCWBR	15.5	22.7	21.4	18.0	15.5	12.9	11.5	13.6	16.5	17.4	11.9	13.1
	5% WB	MCDBR	15.9	25.9	28.7	30.5	28.0	26.2	24.2	27.2	29.3	27.6	15.8	13.3
		MCWBR	14.8	21.5	19.3	16.7	13.5	11.5	10.4	12.4	15.2	16.5	11.5	12.6
Clear-Sky Solar Irradiance	taub	0.294	0.333	0.385	0.434	0.419	0.411	0.426	0.409	0.373	0.367	0.334	0.302	
	taud	2.039	1.975	1.933	2.034	2.185	2.302	2.282	2.299	2.339	2.220	2.075	1.989	
	Ebn at Noon	209	235	246	252	264	267	261	259	255	225	190	177	
	Edh at Noon	25	36	47	48	43	39	39	37	31	29	24	23	
All-Sky Solar Radiation	RadAvg	378	712	1143	1426	1669	1908	1780	1499	1096	599	321	258	
	RadStd	40	53	113	113	172	164	105	108	94	68	40	36	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page